Adaptive Book Imbalance Execution Strategies Lead to Robust Improvements Over Naive Benchmark

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Abstract

A fundamental challenge in medium-frequency systematic trading is the underperformance of simple execution strategies. Naively splitting a parent order into smaller market-orders over seconds or minutes exposes child fills to adverse price drifts, and can also lead to large price impact. In this work we compare the arrival cost of different order splitting and execution strategies from the naive zero-intelligence to the dynamically adaptive, with the goal of reducing the total arrival cost of a parent order, subject to time, quantity, and price constraints. To do this, we use level two orderbook data sourced from Databento, and build an order book simulation capable of simulating the fills and price impact of any arbitrary execution strategy using common order types. We examine three failed execution strategies and their individual advancements and issues. We showcase our final implementation, which achieves robust cost reductions of 55% - 89%, by intelligently calibrating the aggression of child orders to take advantage of favorable price conditions while avoiding unfavorable ones. We then examine this strategy's mechanics, results and statistical significance for three assets and for ranges of latency and book replenishment parameters. Finally, we suggest further tweaks and improvements to the strategy, to be pursued in the future.

1 Motivation and Main Ideas

1.1 Underperformance of Naive Splitting

The main driver of this work is the unfortunate underperformance of simple execution strategies. Naively splitting a parent order into n chunks and filling them as market orders Δt time units apart has high impact and exposes the parent order to significant price drifts. Figure 1 shows the price impact of repeatedly lifting the ask over time. This simulated fill split a 5000-share buy order into twenty-five 200-share fills. The underperformance of this strategy necessitates a more intelligent way to identify favorable ways to fill and split large orders.

1.2 Book Imbalance as a Primary Signal

The core of our improvements over the naive method make use of the imbalance in quantity on the top six levels of the book as a predictive signal over short horizons. Structurally, as marketable buy and sell orders arrive at approximately equal rates, imbalance in top of book depth should predict which levels are depleted fastest and therefore the direction of a subsequent price movement. We chose OBI because it is a well understood, easy to implement, and powerful signal of price direction on the micro-to-milli-second time scale. As evidence of OBI's predictive power, we used the markout plot in Figure 2. Within five separate ranges of OBI values, we sample 100,000 timestamps and calculate the difference between the initial mid-price and the mid-price at 10,000 subsequent times,

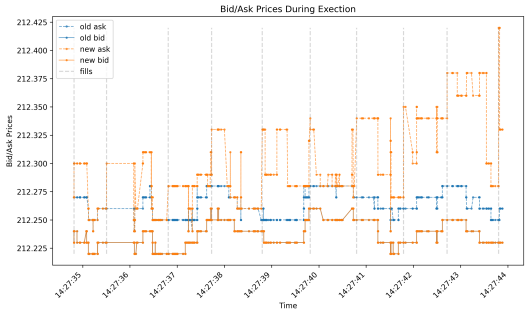


Figure 1: Naive Order Splitting

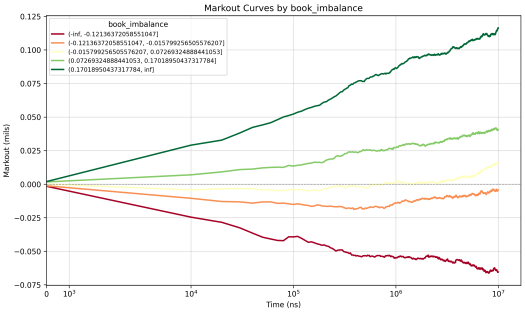


Figure 2: Markout Plot

then average over each of the 100,000 paths. We use a time horizon of $[0, 10]$ milliseconds, in steps of ten microseconds.

Figure 2 clearly shows that OBI is a powerful predictor of price drift within milliseconds and that this price motion is structural, indicated by the length of time that the markouts stay smooth and near-zero variance. Using OBI's predictive qualities, our fill strategy, discussed in Section 2.7 can reliably predict positive and negative price drifts and only fill when advantageous.

2 Methodological Choices and Innovations

2.1 Sourcing Data

Throughout this work we utilize level 2 data in the `MBP-10` schema from Databento, which provides the top 10 levels of the book for a selection of stocks and dates. The dataset samples on all order book updates (trades, adds, cancels, etc.) which allows us to reconstruct the book over time. The data is sourced from Databento. We fetch data through a simple `DatabentoGenericAdapter` class that uses a hierarchical caching structure. This allows us to call the Databento API only for new data, which is necessary to manage time and storage capacity for L2 book data. The adapter caches on `[tickers, start, end, dataset, schema]`. Here, our schema is always `MBP-10` and we use the `XNAS.ITCH` Dataset which gives us full access to the Nasdaq exchange.

We use L2 data for three tickers, `AMZN`, `CSCO`, and `DRH` from `"2026-03-18":"2026-04-18"`. We chose these tickers because they are representative of three major varieties of equities. `AMZN` is a liquid small-tick stock that should offer ample resilience to price impact and relatively high and volatile intraday bid-ask spreads. `CSCO` is a liquid large-tick stock that still has deep liquidity but more narrow and consistent intraday spreads. Finally, `DRH` is an illiquid large-tick stock that offers a challenge to minimizing costs. It cannot accept large amounts of liquidity, moves more slowly than `AMZN` and `CSCO`. It also has more inconsistent order arrival rates, which poses a direct challenge for using OBI as a signal. Altogether this dataset is 28.7GB, on the order of RAM capacity for a sequential process but requires some management for parallel processing tasks, like the below.

2.2 Simulator

The technical backbone of this work is the `ExecutionSimulator` class we designed to simulate fills using raw `MBP-10` data. The simulator can do the following.

1. Simulate the execution of market orders, resting limit orders, and immediate-or-cancel (IOC) limit orders at a given nanosecond timestamp and latency.
2. Maintain an orderbook state over time, using level two `MBP-10` data from Databento, consistent with mutations of the book caused by simulated fills and the replenishment of swept levels via a Poisson process with book-update sampling.
3. Calculate and track execution statistics (arrival cost, fill times, quantities, etc.) for any arbitrary execution strategy using the above order types.

Our implementation of the `ExecutionSimulator` class initiates the L2 book with a `polars` dataframe in Databento's `MBP-10` schema. The simulator maintains the book using internally consistent `numpy` arrays, which it updates as the book mutates via simulated fills. After each simulated order is filled (mainly child orders) the simulator returns a dictionary of the form below. Note from Section 5 that `arrival_cost` is calculated as the `vwap` minus the mid price at execution. At the end

of a simulation it returns a new `polars` dataframe representing the mutated book over time (sampled at book-update events) as well as a list of individual fill results described in Section 5.

At each book update the simulator also replenishes any levels swept by previous simulated fills. After each marketable order execution, the simulator draws a limit order quantity q_i for each price tick p_i between the best ask and best bid, using a Poisson distribution at rate λ (set as a hyperparameter). If any $q_i > 0$, then a limit order is added with quantity q_i at price level p_i.

2.3 Strategy Evaluation

To evaluate each of the following strategies, we utilize a parallel processing trial workflow. At initialization, we set the numerous required hyperparameters, each listed below.

- `Latency`: 100us, `Poisson Replenishment`: 100.0, `PARENT_QTY`: 10_000, `MAX_WORKERS`: 10, `N_TRIALS`: 5_000, `SEED`: 12345, `HORIZON`: 300s

To load data while managing memory we scan the full dataset's cache and split into one `.parquet` for each ticker. We then randomly sample `N_TRIALS` initial timestamps from the book, ensuring that no two windows from the initial timestamp through the `HORIZON` overlap more than a few seconds. Then for each combination of asset and fill strategy, we initialize `MAX_WORKERS` CPU workers and scan the relevant ticker `.parquet` file for each window and run the fill strategy on that window only, randomly sampling the side of the order from `['BUY', 'SELL']`. This workflow balances memory requirements and parallelized processing speed. Compared with the sequential workflow used in the milestone report, this brought down the time cost from half an hour per trial over the full dataset to a few seconds per trial over a selected window.

To measure the performance of each fill strategy, we save information including `side`, `parent_filled`, `parent_arrival_cost`, `mean_child_arrival_cost`, `median_child_arrival_cost`, and `parent_lifetime`. Below we compare our final successful strategy with the naive strategy using the above metrics, and go over some failed innovations and their associated learnings along the way. More on specific definitions of the above metrics can be found in Section 5.

2.4 Naive Benchmark

As discussed and plotted earlier, we are comparing our strategies with the naive method of splitting orders into `N_CHUNKS = 50`, and filling them a fixed time interval `WAIT_TIME = 1s` apart using regular market orders. Table 1 shows some performance metrics for this naive method filling ten-thousand shares of `CSCO` to serve as a benchmark we are competing against.

Table 1: Naive Results

statistic str	total_filled f64	parent_cost f64	mean_child_cost f64	median_child_arrival_cost f64	lifetime_s f64
"mean"	10000.0	9.8626	5.798219	4.182775	9.582253
"min"	10000.0	-21.535187	1.374581	0.580619	6.221948
"25%"	10000.0	6.428417	3.883612	3.316627	9.524487
"50%"	10000.0	9.053304	5.149228	3.90504	9.687721
"75%"	10000.0	12.472484	7.069212	4.637834	9.761385
"max"	10000.0	61.71924	36.425326	35.814917	9.800094

2.5 Failure: Naive with IOC

The first pass attempt to improve on the benchmark is simply running the same strategy using marketable IOC limit orders instead of market orders. Results below in Table 2. We set the limit price of each child order to the 10th level of the ask (bid) for a parent buy (sell) order at the initial time of execution. This resulted in better fills, however it almost never completed the full parent order given its simple cap on the allowed execution price.

Table 2: Naive IOC Results

statistic str	total_filled f64	parent_cost f64	mean_child_cost f64	median_child_arrival_cost f64	lifetime_s f64
"mean"	7818.5352	5.28774	3.374856	3.293796	8.880517
"min"	200.0	-21.525161	1.207293	0.580619	0.0
"25%"	6838.0	4.482656	3.052005	2.950195	8.446878
"50%"	7905.0	5.518874	3.353437	3.200819	9.370963
"75%"	8967.0	6.410203	3.658963	3.726398	9.709241
"max"	10000.0	29.237672	16.116962	15.40023	9.800094

2.6 Failure: OBI Aggressive Only

Beyond simply tweaking the benchmark our first true pass at an improvement used OBI to time our fills using the parameter `OBI_THRESHOLD = 0.85`. First, set the maximum size of a child fill at q_{max}. For a buy order, if the OBI was above the threshold that indicated an expected upward price move, the logic is to fill a child order immediately of size $q_{max} * |OBI|$, where $q_{max} = 350$. This would dynamically size orders based on the size of the subsequent price moves, and attempt to only fill at advantageous times. Figure 3 shows this strategy in action.

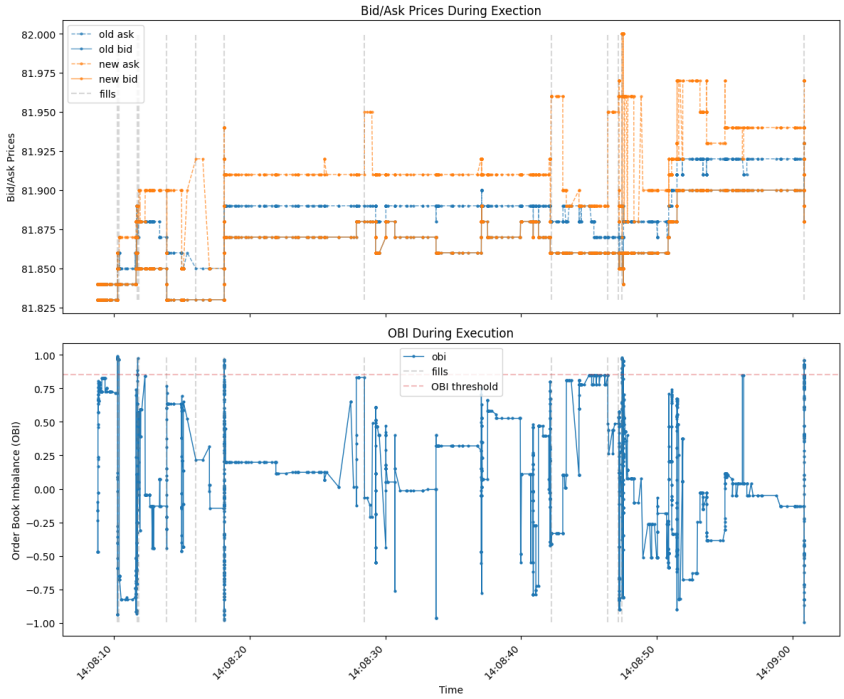


Figure 3: OBI Aggressive Only Fills

This strategy has a few issues. Firstly, for some time horizons, OBI infrequently or simply never hits the threshold and so the parent order only gets a partial fill. More fundamentally, this logic only seeks to avoid adverse fill conditions, and does nothing explicit to take advantage of favorable ones.

The aggressive-only OBI logic originated from a way of thinking rooted in alpha generation and the predicted returns to be exploited by taking an aggressive position. But returns are exactly *not* what an execution strategy is focused on: simply put, to take advantage of alpha a strategy must enter and exit a position, but an execution strategy is *given* a predetermined order to place and its task is to minimize the cost filling the order's quantity over a set time horizon. Solving the second issue via this change of perspective lead directly to our improvement via a hybrid strategy.

2.7 Final Strategy: Imbalance Calibrated Aggression

After escaping the alpha generation line of thinking, we were able to develop a much more effective fill strategy. It is a hybrid of the naive and OBI strategies that uses OBI to calibrate the aggressiveness or urgency of fills. For a parent order of quantity q logic is as follows:

1. Split the parent order into n child orders of size q/n.
2. Set positive and negative OBI thresholds for expected price moves that are calibrated using the markout plots above.
3. For a buy order, depending on the value of OBI, do the following (or opposite for a sell):

- If OBI > `OBI_THRESHOLD`, expect an adverse price move. Fill one child order immediately. Cross the spread with an IOC limit order and limit price set to the 10th level of the book at the time the parent order was split.
- If OBI < -1 * `OBI_THRESHOLD`, expect a favorable price move. Place resting limit orders at each of the top ten bids, each for the quantity of one child order. Set expiry at the end of the parent order's allowed horizon.
- If OBI is between the above thresholds, fill child orders with IOC limit orders at a slow constant rate with the same limit price.

By calibrating the aggression of child orders and allowing the price to drift, this strategy fills the vast majority of shares during favorable price conditions, and avoids unfavorable ones by filling immediately and then not at all until the price moves favorably again. This is illustrated in Figure 5c, which shows the true and simulated best bid and ask over time with favorable and adverse price drifts highlighted using the value of OBI. The vertical gray markers are the timestamps of each child fill. Note that limit orders placed when OBI is low may fill when OBI is high, resulting in lots of fills during times marked "unfavorable". This is positive and expected behavior.

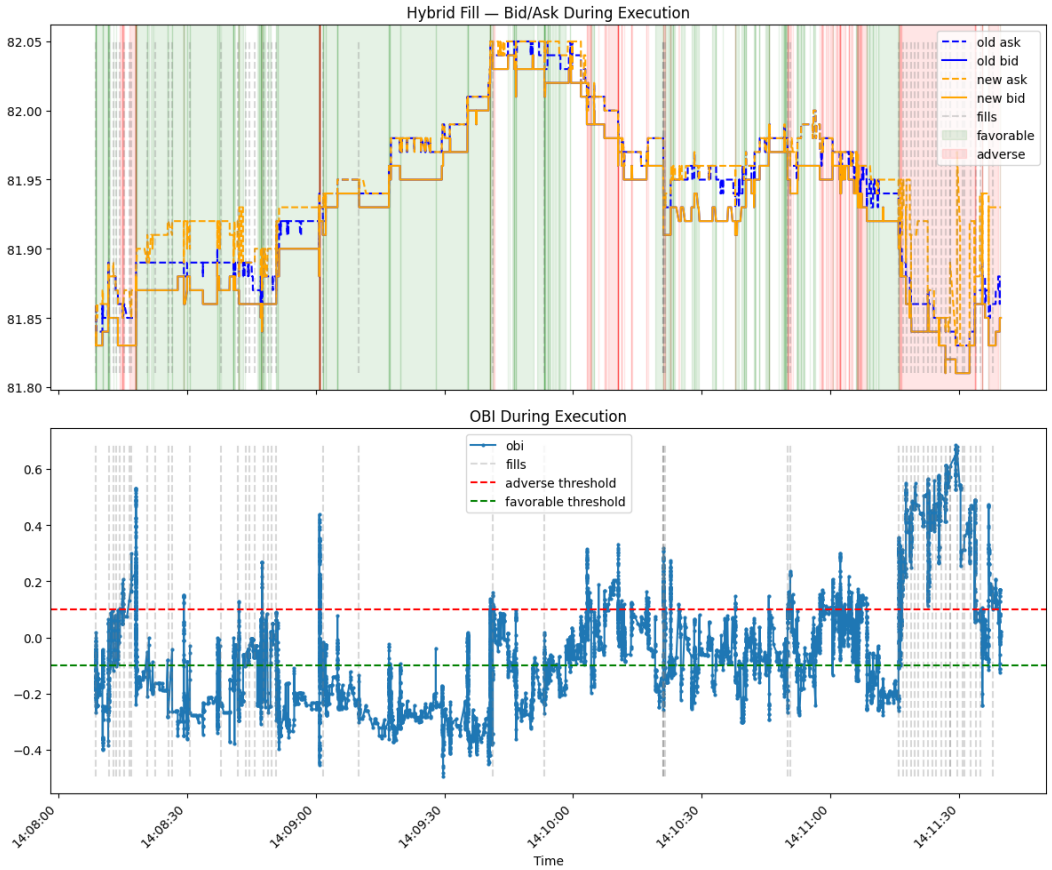


Figure 4: Hybrid Fill Strategy

Table 3: Hybrid Fill Statistics

(a) AMZN

statistic	parent_cost	lifetime_s
str	f64	f64
"mean"	0.184687	28.482708
"std"	2.185917	45.185725
"50%"	0.069449	13.0227

(b) CSCO

statistic	parent_cost	lifetime_s
str	f64	f64
"mean"	1.496204	32.403243
"std"	4.217447	49.10376
"50%"	2.133194	15.597951

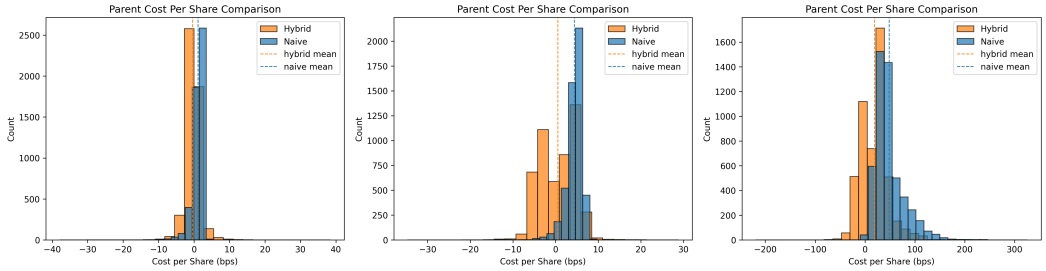
(c) DRH

statistic	parent_cost	lifetime_s
str	f64	f64
"mean"	25.47164	75.61788
"std"	28.592753	92.533696
"50%"	28.342588	31.240362

3 Empirical Findings

Table 3 shows some descriptive statistics of the results of the hybrid fill strategy, by asset, and Table 4 in Section 3.1 shows the means of the naive and hybrid strategies alongside their statistical significance demonstrated by a t-test. Parent costs are measured in bps per share. These results are derived from running the strategy across assets for 5000 timestamps, latency of `100us`, replenishment rate of $\lambda = 15$ (which is quite low), a time horizon of 5 minutes, and splitting a 10,000 share parent order into $n = 50$ child orders each with quantity $q = 200$ with OBI thresholds of ± 0.10.

The average parent order's total cost, defined in Section 5, is highly dependent on the liquidity of each stock, with `DRH` and `CSCO` fills being respectively 141x and 9x times more costly than `AMZN` on average. This is likely the result of illiquidity causing child orders to have higher mechanical price impact, and methods to improve this are considered in Section 4. The order lifetimes are relatively consistent at approximately thirty seconds for the liquid stocks, and above a minute for illiquid. All hybrid fills were completed within the 5 minute horizon. Figure 5 shows histograms of the total parent cost per asset. Here it is visually obvious that the naive method has a heavy right tail, while the hybrid strategy has a mean shift downward and heavy left tail. Its ability to deliver these results with such a low replenishment rate simulating small sub-100-share limit order replenishment is impressive.



(a) AMZN (b) CSCO (c) DRH

Figure 5: Total Parent Cost Comparison

3.1 Test Stats / Significance

Table 4 shows the average cost in bps per share, per ticker, using the hybrid and naive fill strategies, as well as the results of a paired T-test sampled per asset. The hybrid strategy shows a pronounced downward shift in cost across assets, with improvements of 89% for `AMZN`, 73% for `CSCO`, and 55% for `DRH`. Each of these results is also highly significant. We chose to use a paired t-test since each strategy was run on the same 5,000 time windows, and therefore an independent t-test would be the wrong choice.

Table 4: Paired T-Test on Arrival Cost per Asset

symbol str	hybrid mean f64	naive mean f64	t_stat f64	p_val f64
"AMZN"	0.18	1.59	33.74	0.0
"CSCO"	1.5	5.24	56.09	0.0
"DRH"	25.47	54.43	50.81	0.0

3.2 Effects of Latency and Replenishment

In Table 5 and Table 6 we demonstrate the hybrid strategy's robustness to latency and replenishment. Here, latency is implemented as a delay between the time each child order is sent and when it enters the book, and is displayed in microseconds. The bps per-share cost differences between levels of latency are remarkably low across assets and strategies. This is obvious for the naive strategy as it is zero-intelligence. The hybrid strategy does have some reliance on latency when filling with an IOC order immediately before adverse moves, however the majority of fills are from resting limit orders filling over multiple seconds and so it remains robust to changes in latency, even up to fifty milliseconds.

The effects of low replenishment are also small but are more clear than latency, while both strategies benefit heavily from a high rate of replenishment. Across strategy and asset, a higher rate $\lambda = 200$ leads to a lower price impact and thus lower average costs. This is simply calibration of the book's liquidity, and demonstrates a sanity check on the fill results.

Table 5: Effects of Latency

symbol str	latency_level str	hybrid mean cost f64	naive mean cost f64	latency (us) f64
"CSCO"	"low"	4.19	9.76	10.0
"AMZN"	"low"	1.36	4.3	10.0
"DRH"	"low"	35.53	88.32	10.0
"CSCO"	"mid"	4.19	9.86	100.0
"AMZN"	"mid"	1.42	4.17	100.0
"DRH"	"mid"	35.62	88.36	100.0
"CSCO"	"high"	4.38	10.27	50000.0
"AMZN"	"high"	1.35	4.37	50000.0
"DRH"	"high"	35.64	88.54	50000.0

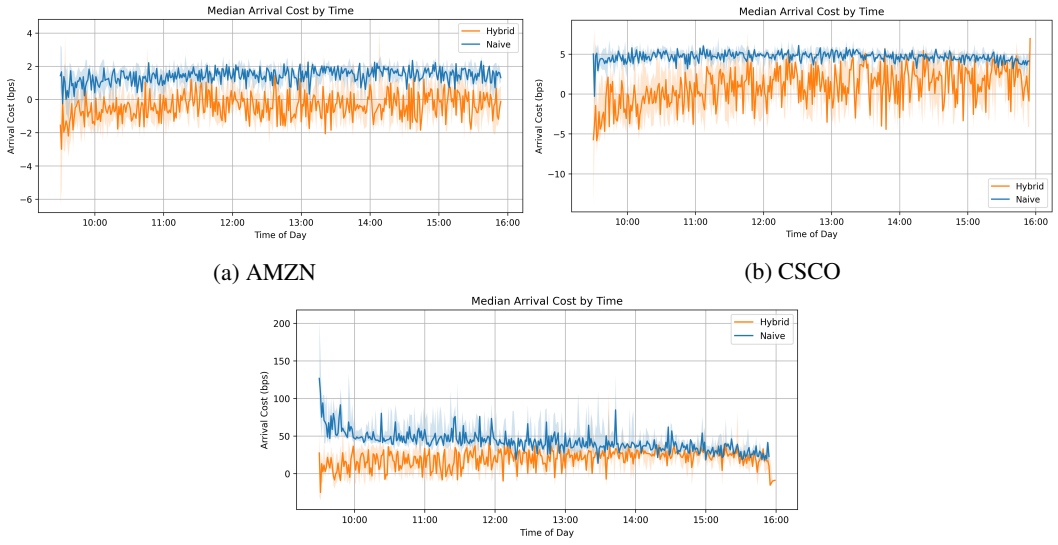
Table 6: Effects of Replenishment

symbol str	replenishment_level str	hybrid mean cost f64	naive mean cost f64	replenishment f64
"CSCO"	"low"	5.37	13.63	1.0
"AMZN"	"low"	2.03	6.9	1.0
"DRH"	"low"	42.01	94.8	1.0
"CSCO"	"mid"	4.19	9.86	15.0
"AMZN"	"mid"	1.42	4.17	15.0
"DRH"	"mid"	35.62	88.36	15.0
"CSCO"	"high"	-1.51	2.62	200.0
"AMZN"	"high"	-1.38	0.38	200.0
"DRH"	"high"	14.48	38.46	200.0

3.3 Intraday Seasonality

Figure 6 shows the intraday seasonality of median arrival cost in bps per-share, across tickers and strategies. As expected both strategies, though especially the hybrid, benefit from the high volume on the open, and approach a mean throughout the day. Interestingly, the naive strategy does worst at the

open for DRH. This is likely a result of both high opening spreads and low top book depth on open for the illiquid asset, which then stabilize over time. This effect is demonstrated in Lecture 8.



(a) AMZN

(b) CSCO

(c) DRH

Figure 6: Intraday Seasonality of Arrival Cost

4 Future Work

Overall, the imbalance calibrated aggression strategy is a highly effective one. As always though, a few tweaks stand out in their potential to improve upon these results. Each is laid out below.

4.1 EMA of Book Imbalance

To prevent temporary peaks in book imbalance from affecting the strategy's aggression, the strategy could use a moving average of book imbalance. This would smooth the peaky nature of OBI. However, this introduces yet more hyperparameters and is linked with the time to expiry of resting limit orders.

4.2 Catch-Up Mechanism

Currently, the strategy is placing naive IOC orders when OBI is not strongly indicating the direction of price drift, which can cause unnecessary price impact. This could be improved by implementing a mechanism to measure how ahead or behind the strategy is in its filled quantity and dynamically tweak the OBI thresholds or size of fills.

4.3 Horizon as a Function of Volume

Currently the hybrid strategy uses a set hyperparameter for the parent order's horizon to fill. This could be dynamically chosen as a function of the share or dollar volume in the past few minutes, indicating how much impact the fills are likely to have. With more volume being traded, the time horizon can be shorter and vice-versa.

4.4 Improving Book Simulation

Calibrating hyperparameters like the poisson rate of replenishment would help achieve more accurate results. Access to proprietary trading data would also inform the choice of distribution for limit order replenishment and sizes, as a Poisson distribution is a relatively simple approximation.

5 Definitions and Appendix

Throughout this work we refer to a few concepts that require specific definitions. We lay them out below for reference.

- OBI: We define order book imbalance (OBI) using the top *six* levels of the book, as follows:

$$OBI = \frac{\Sigma_{i=0}^{5}[Q_i^{bid} - Q_i^{ask}]}{\Sigma_{i=0}^{5}[Q_i^{bid} + Q_i^{ask}]} \in [-1, 1] \tag{1}$$

- Price Impact : When referring to price impact we include only the mechanical impact of taking liquidity from the limit order book. Our order book simulation is discussed in Section 2.2, and includes further information on impact as well as replenishment and latency.
- Arrival Cost: When discussing to the arrival cost of a parent or child order, we are referring to the volume-weighted average price of a fill minus the mid price at the time of execution, divided by mid price and converted to bps. For a parent order, the VWAP sums over child order's individual vwaps in the same way, and the mid price is set to the price at the time the parent order was sent to the execution engine. Below, q_i and p_i are the individual fill's quantities and prices respectively.

$$\begin{aligned} p_{vwap} &= \frac{\Sigma_{i=0}^{n} q_i \, p_i}{\Sigma_{i=0}^{n} q_i} \\ \text{ArrivalCost (bps)} &= \frac{p_{vwap} - p_{mid}}{p_{mid}} * 10^4 \end{aligned} \tag{2}$$

Discussed in Section 2.2, each simulated individual fill returns a dictionary of fill statistics, below.

```
{'vwap': 211.9365352112676,
 'fills': [(211.93, 246), (211.94, 464)],
 'total_filled': 710,
 'levels_swept': 1,
 'executed_at_ns': 1774538471626080001,
 'arrival_cost': 0.02153521126757596}
```

6 References

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